

Small Networks, Solid Baselines: An MLP Study on Handwritten Digits

Abstract

We revisit simple baselines for handwritten digit recognition. A one-hidden-layer multilayer perceptron reaches 97.2% test accuracy on the scikit-learn digits dataset, outperforming logistic regression. We release code for all experiments.

1 Introduction

Prior work using kernel methods reported 98.9% accuracy on a related benchmark [3]. We focus on models that train in seconds on a laptop CPU.

2 Experimental setup

We use the scikit-learn digits dataset (1,797 images of 8x8 pixels, 10 classes). Pixel values are scaled to [0, 1] by dividing by 16. We use an 80/20 train/test split. The MLP has 64 hidden units with ReLU activations and is trained with Adam at a learning rate of 0.001 for 200 epochs. Logistic regression uses the lbfgs solver.

3 Results

Table 1: Test results on digits (mean over 5 runs).

Model	Accuracy (%)	Macro-F1
Logistic regression	96.1 $\pm$ 0.3	0.960
MLP (64 hidden)	97.2 $\pm$ 0.4	0.971
MLP (128 hidden)	97.5 $\pm$ 0.3	0.974

4 Discussion

Doubling the hidden layer to 128 units gives a small gain of 0.3 points.

Training the MLP takes under 10 seconds on a single CPU core.

References

[3] A. Author. Kernel methods for digits. 2015.